

Guideline for the pharmacological treatment of hypertension in adults



World Health
Organization

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Acronyms and abbreviations

ACE1	angiotensin-converting enzyme 1
ACE2	angiotensin-converting enzyme 2
ACEi	angiotensin-converting enzyme inhibitor
ARB	angiotensin-II-receptor blocker
BB	beta-blocker
BP	blood pressure
CAD	coronary artery disease
CCB	calcium channel blocker
CKD	chronic kidney disease
CRE	WHO Office of Compliance, Risk Management and Ethics
CV	cardiovascular
CVD	cardiovascular disease
DBP	diastolic blood pressure
DM	diabetes mellitus
DOI	declaration of interest
ECG	electrocardiogram
EML	Essential Medicines List
ERG	External Review Group
ESRD	end-stage renal disease
GDG	Guideline Development Group
GRADE	Grading of Recommendations Assessment, Development and Evaluation
HCW	health care worker (nonphysician)
HIC	high-income country
HTN	hypertension
LIC	low-income country
LMIC	low- and middle-income country
LVH	left ventricular hypertrophy
MACE	major adverse cardiovascular event
MI	myocardial infarction
MIC	middle-income country
NCD	noncommunicable disease
PEN	WHO package of essential NCD interventions
PICO	population intervention comparator outcome
QALY	quality-adjusted life year
RAAS	renin-angiotensin-aldosterone system
SBP	systolic blood pressure

Executive summary

More people die each year from cardiovascular diseases than from any other cause. Over three quarters of heart disease and stroke-related deaths occur in low-income and middle-income countries. Hypertension – or elevated blood pressure – is a serious medical condition that significantly increases the risk of heart, brain, kidney and other diseases. Hypertension can be defined using specific systolic and diastolic blood pressure levels or reported use of antihypertensive medications. An estimated 1.4 billion people worldwide have high blood pressure, but just 14% have it under control. However, cost-effective treatment options do exist.

In this guideline, the World Health Organization (WHO) provides the most current and relevant evidence-based global public health guidance on the initiation of treatment with pharmacological agents for hypertension in adults. The recommendations target adult, non-pregnant patients who were appropriately diagnosed with hypertension and counselled about life-style modifications.

The guideline provides new recommendations on the threshold for the initiation of pharmacological treatment for hypertension, as well as recommendations on intervals for follow up, target blood pressure to be achieved for control, and the cadre of health care workers who may initiate treatment. The guideline provides the basis for deciding whether to initiate treatment with monotherapy, dual therapy or single-pill combinations, as well as guidance for countries selecting medicines and algorithms for hypertension control for their national guidelines for hypertension management.

The guideline was developed in accordance with the WHO Handbook for Guideline Development. In brief, the WHO Steering Group, in collaboration with the Guideline Development Group, developed key questions and rated outcomes to identify those critical for the development of the guideline. Conflicts of interest were handled in line with the current Compliance, Risk Management and Ethics (CRE) policy and all members of the GDG were asked to fill in the standard WHO Declaration of Interest (DOI) forms, which were reviewed. An overview of systematic reviews of the evidence was used to build summary of findings tables according to the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) approach. The Guideline Development Group developed recommendations, considering the certainty of the evidence; the balance between desirable and undesirable effects; resource requirements and cost-effectiveness; health equity; acceptability, patient values and preferences, and feasibility.

Recommendations

1. RECOMMENDATION ON BLOOD PRESSURE THRESHOLD FOR INITIATION OF PHARMACOLOGICAL TREATMENT

WHO recommends initiation of pharmacological antihypertensive treatment of individuals with a confirmed diagnosis of hypertension and systolic blood pressure of ≥ 140 mmHg or diastolic blood pressure of ≥ 90 mmHg.

Strong recommendation, moderate- to high-certainty evidence

WHO recommends pharmacological antihypertensive treatment of individuals with existing cardiovascular disease and systolic blood pressure of 130–139 mmHg.

Strong recommendation, moderate- to high-certainty evidence

WHO suggests pharmacological antihypertensive treatment of individuals without cardiovascular disease but with high cardiovascular risk, diabetes mellitus, or chronic kidney disease, and systolic blood pressure of 130–139 mmHg.

Conditional recommendation, moderate- to high-certainty evidence

2. RECOMMENDATION ON LABORATORY TESTING

When starting pharmacological therapy for hypertension, WHO suggests obtaining tests to screen for comorbidities and secondary hypertension, but only when testing does not delay or impede starting treatment.

Conditional recommendation, low-certainty evidence

3. RECOMMENDATION ON CARDIOVASCULAR DISEASE RISK ASSESSMENT

WHO suggests cardiovascular disease risk assessment at or after the initiation of pharmacological treatment for hypertension, but only where this is feasible and does not delay treatment.

Conditional recommendation, low-certainty evidence

4. RECOMMENDATION ON DRUG CLASSES TO BE USED AS FIRST-LINE AGENTS

For adults with hypertension requiring pharmacological treatment, WHO recommends the use of drugs from any of the following three classes of pharmacological antihypertensive medications as an initial treatment:

1. thiazide and thiazide-like agents
2. angiotensin-converting enzyme inhibitors (ACEis)/angiotensin-receptor blockers (ARBs)
3. long-acting dihydropyridine calcium channel blockers (CCBs).

Strong recommendation, high-certainty evidence

5. RECOMMENDATION ON COMBINATION THERAPY

For adults with hypertension requiring pharmacological treatment, WHO suggests combination therapy, preferably with a single-pill combination (to improve adherence and persistence), as an initial treatment. Antihypertensive medications used in combination therapy should be chosen from the following three drug classes: diuretics (thiazide or thiazide-like), angiotensin-converting enzyme inhibitors (ACEis)/angiotensin-receptor blockers (ARBs), and long-acting dihydropyridine calcium channel blockers (CCBs).

Conditional recommendation, moderate-certainty evidence

6. RECOMMENDATIONS ON TARGET BLOOD PRESSURE

WHO recommends a target blood pressure treatment goal of <140/90 mmHg in all patients with hypertension without comorbidities.

Strong recommendation, moderate-certainty evidence

WHO recommends a target systolic blood pressure treatment goal of <130 mmHg in patients with hypertension and known cardiovascular disease (CVD).

Strong recommendation, moderate-certainty evidence

WHO suggests a target systolic blood pressure treatment goal of <130 mmHg in high-risk patients with hypertension (those with high CVD risk, diabetes mellitus, chronic kidney disease).

Conditional recommendation, moderate-certainty evidence

7. RECOMMENDATIONS ON FREQUENCY OF ASSESSMENT

WHO suggests a monthly follow up after initiation or a change in antihypertensive medications until patients reach target.

Conditional recommendation, low-certainty evidence

WHO suggests a follow up every 3–6 months for patients whose blood pressure is under control.

Conditional recommendation, low-certainty evidence

8. RECOMMENDATION ON TREATMENT BY NONPHYSICIAN PROFESSIONALS

WHO suggests that pharmacological treatment of hypertension can be provided by nonphysician professionals such as pharmacists and nurses, as long as the following conditions are met: proper training, prescribing authority, specific management protocols and physician oversight.

Conditional recommendation, low-certainty evidence

1 Introduction

More people die each year from cardiovascular diseases than from any other cause. Over three quarters of heart disease and stroke-related deaths occur in low-income and middle-income countries (1). Blood pressure is the force exerted by circulating blood against the walls of the body's arteries, the major blood vessels in the body. Blood pressure is written as two numbers. The first number (systolic) represents the pressure in blood vessels when the heart contracts or beats. The second number (diastolic) represents the pressure in the vessels when the heart rests between beats. Hypertension – or elevated blood pressure – is a serious medical condition that significantly increases the risk of diseases of the heart, brain, kidneys and other organs (2). Hypertension can be defined using specific systolic and diastolic blood pressure levels or reported use of antihypertensive medications. An estimated 1.4 billion people worldwide have high blood pressure, but just 14% have it under control (2). However, cost-effective treatment options do exist.

A World Health Organization (WHO) guideline dealing specifically with raised blood pressure was last published more than 20 years ago – in 1999 – and is now outdated. In 2007, a comprehensive guideline on cardiovascular risk included some recommendations on hypertension but this also needs revision and updating in the light of new evidence and practice (3). Guidance is particularly needed on some controversial issues, such as when to start treatment and whether laboratory testing and cardiovascular risk assessment are needed prior to starting treatment. In the past decade, WHO has included diagnosis and management of hypertension in a total cardiovascular risk approach as part of the WHO package of essential noncommunicable disease (NCD) Interventions (WHO PEN) 2007, 2010 and 2013. However, this approach has not included the most recent advances in pharmacological treatment.

The WHO Essential Medicines List (EML) identifies all classes of antihypertensive drugs – angiotensin-converting enzyme inhibitors (ACEi), calcium channel blockers (CCB), angiotensin receptor blockers (ARB) and thiazide diuretics – as essential. In June 2019, the EML included single-pill combination medications for hypertension. This further supports the evaluation of all classes of antihypertensive drugs as well as single-pill combinations in this current guideline.

Scope and objectives of the hypertension guideline

The 2021 WHO hypertension guideline aims to provide the most current and relevant evidence-based global public health guidance on the initiation of treatment (with pharmacological agents) for hypertension in adults. The recommendations target the general adult, non-pregnant, hypertensive population.

Although several countries and professional societies have guidelines on the topic of hypertension, these are specific to the population of that particular country or the specific setting or constituency of the professional society. Recent shifts in hypertension management, such as moving away from using beta-blockers as a first-line agent or the increased research and adoption of combination therapies and single-pill combinations, are all additional reasons for new guidance. The *Guideline for the pharmacological treatment of hypertension in adults* will be the first global guideline in the past two decades on the topic and will have specific relevance to low- and middle-income countries (LMICs).

The guideline provides new recommendations on the threshold for the initiation of pharmacological treatment for hypertension, recommendations on intervals for follow up, target blood pressure to be achieved for control, and the cadre of health care workers who may initiate treatment. It provides the basis for deciding whether to initiate treatment with monotherapy, dual therapy, or single-pill combination, as well as guidance for countries on selecting medicines for hypertension control in their national guidelines for hypertension management.

This guideline will replace the guidance in the modules *Evidence-based protocols* and *Risk-based CVD management* of the HEARTS technical package, as well as the guidance in the WHO PEN package regarding the threshold for initiation of treatment, and for the preferred pharmacological treatment for hypertension.

The guideline does not address measurement of blood pressure or diagnosis of hypertension. It addresses pharmacotherapy in individuals with a “confirmed” diagnosis of hypertension, such as hypertension diagnosed when blood pressure is found to be elevated on two different days.

Although this guideline does not address modifiable risk factors for hypertension such as unhealthy diet, physical inactivity, consumption of tobacco and alcohol, and being overweight or obese, a comprehensive treatment plan for hypertension must include addressing these risk factors through life-style modifications and other interventions (2).

Nonpharmacological approaches to treatment or prevention of hypertension include:

- reducing salt intake (to less than 5g daily)
- eating more fruit and vegetables
- being physically active on a regular basis
- avoiding use of tobacco
- reducing alcohol consumption
- limiting the intake of foods high in saturated fats
- eliminating/reducing trans fats in diet (2).

The guideline does not address hypertensive crisis as it is focused on chronic blood pressure management in regular care settings.

The objectives of the hypertension guideline are to:

- provide a blood pressure threshold for the initiation of treatment for hypertension;
- determine if laboratory tests or cardiovascular risk assessment are required prior to initiation of treatment for hypertension;
- determine the pharmacological agents with which to initiate treatment;
- determine the need to initiate treatment with monotherapy, dual therapy, or single-pill combinations;
- provide targets for blood pressure control in hypertension;
- provide intervals for follow up for patients with hypertension; and
- determine how nonphysician health care workers can participate in the management of hypertension.

Target audience

Primary audience

Clinicians/health care providers at all levels of health care.

Secondary audience

National NCD/CVD programme managers, health care academics, policy-makers setting practice recommendations, students, and hypertension medicine manufacturers.

2 Method for developing the guideline

2.1 Guideline contributors

In order to develop the hypertension guideline, WHO established three groups:

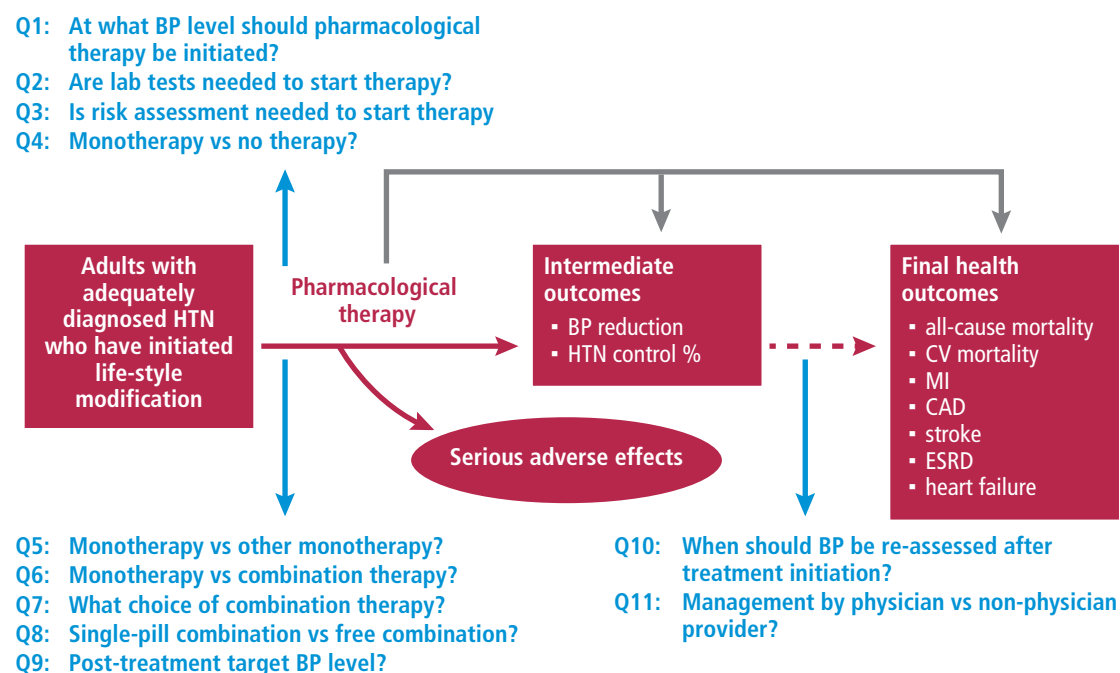
1. An internal WHO Steering Group to coordinate the guideline development process.
2. A Guideline Development Group (GDG), composed of hypertension (HTN) physicians, nephrologists, cardiologists, pharmacists, nurses, researchers, academics, policy-makers and representatives of patient groups, to review the evidence and develop recommendations. WHO selected the members of the GDG based on relevant expertise but it also considered appropriate representation by region and sex.
3. An External Review Group (ERG), composed of technical experts, representatives of HTN patient groups and ministries of health from low-resource countries, to provide peer review of the guideline and ensure recommendations are aligned with current global needs.

Annex 1 lists the contributors in each group. Annex 2 describes the process for declaring and managing conflicts of interest.

2.2 Analytical framework and PICOs

An initial GDG meeting was held in Geneva in July 2019 to determine the scope and PICO (population, intervention, comparison, outcome) questions of the guideline. The GDG first developed an analytical framework (Fig. 1) that demonstrates the impact of interventions on intermediate and final outcomes and displays the order of the key questions to better visualize and place them along the patient-flow pathway. Following this, and a preliminary scoping review and discussion between the steering group and methodologist, PICO questions were developed. These were considered, deliberated on, further refined and voted on during the meeting.

Fig. 1 Analytic framework for antihypertensive medication treatment



2.3 Outcome importance rating

Members of the WHO Steering Group, in consultation with the GDG and methodologist, developed a list of treatment outcomes most relevant to the care of individuals with HTN. The GDG then rated each outcome on a scale from 1 to 9 and indicated whether it considered each outcome critical (rated 7–9), important (rated 4–6) or not important (rated 1–3) for decision-making. The mean scores are provided in Annex 3.

2.4 Reviews of evidence

The WHO Steering Group, with the participation of the GDG, determined the scope of the guideline and identified eleven questions in the format of population, intervention, comparison, and outcomes (PICO) to guide the search for systematic reviews (Annex 4). Eleven overviews of reviews informed the guideline development process. A systematic search was carried out in PubMed, Embase, The Cochrane Library, and Epistemonikos to identify existing systematic reviews that answered the PICO questions published in 2015 or after. Suitable systematic reviews were then evaluated based on the following criteria:

- their methodology as appraised by the AMSTAR (Assessing the Methodological Quality of Systematic Reviews) tool;
- how directly they matched the PICO questions;
- whether they reported sufficient information to allow for an assessment of the certainty of the evidence (e.g. tables with characteristics of included studies, risk of bias assessments at the study level, results of meta analyses in forest plots);
- whether they reported evidence in subgroups of interest (e.g. patients with diabetes mellitus (DM), cardiovascular disease (CVD), chronic kidney disease (CKD) etc); and
- the date of the most recent review to ensure the most updated evidence was used.

The Systematic Review Team prioritized the most useful reviews for each question, comparison, outcome and subpopulation within questions, and included as many reviews as necessary to address each question comprehensively. (Web Annex A provides details of the search terms and strategy used.) No reviews were updated.

Two questions (PICO 2 and 10, see Annex 4) were not addressed in an existing systematic review, and evidence from primary studies was therefore used. In this case, the Systematic Review Team reviewed the list of studies used in existing guidelines, queried the GDG and content experts, and searched for primary studies.

A total of 159 systematic reviews and 17 additional primary studies were included. Most of these were traditional systematic reviews in which the authors conducted pairwise meta-analysis, whereas nine analysed individual patient data. The Systematic Review Team also identified several published network meta-analyses. (See Web Annex A for full details.)

Most of the included reviews were found to be of high certainty when assessed using the AMSTAR tool. Details of the selection process, and the reviews and studies included for each PICO question, are presented in Web Annex A.

Another overview of reviews was conducted to identify systematic reviews to inform other decision criteria in the evidence-to-decision framework, including people's values, resources, acceptability, feasibility and equity, presented in Web Annex A. This review was enriched by primary studies identified by GDG members. This review focused on evidence informing HTN management in low- and middle-income settings.

2.5 Certainty of evidence and strength of recommendations

The GDG rated the certainty of evidence and developed the recommendations using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach (4). When making recommendations, GRADE defines the certainty of a body of evidence as “the extent of our confidence that the estimates of an effect are adequate to support a particular decision or recommendation” (5).

Members of the GDG, with the help of the methodologist, developed evidence profiles to summarize relative and absolute estimates of effects, and an assessment of the certainty of the evidence. One evidence profile for each comparison within a PICO question was constructed, using the online Guideline Development Tool GRADEpro (<https://grade.pro.org>). When there were systematic reviews addressing the relative effects in specific subpopulations, separate evidence profiles were constructed for each subpopulation.

According to the GRADE approach, the certainty of the evidence can be high, moderate, low, or very low. Bodies of evidence from randomized controlled trials – which comprised almost the totality of those included in this guideline – start the assessment as high-certainty evidence but can be down-rated due to considerations of risk of bias, inconsistency, imprecision, indirectness, and publication bias. Table 1 presents an explanation of the different levels of certainty of the evidence (5).

Table 1 Certainty of evidence and its implications

Certainty level	Definition
High	We are very confident that the true effect lies close to that of the estimate of effect.
Moderate	We are moderately confident in the effect estimate. (The true effect is likely to be close to the estimate of effect, but there is a possibility that it is substantially different.)
Low	Our confidence in the effect estimate is limited. (The true effect may be substantially different from the estimate of the effect.)
Very low	We have very little confidence in the effect estimate. (The true effect is likely to be substantially different from the estimate of effect.)

The strength of the recommendations reflects the degree of confidence of the GDG that the desirable effects (e.g. beneficial health outcomes) of the recommendations outweigh the undesirable effects (e.g. adverse effects). The strength of recommendations in this guideline was graded into two categories:

A strong recommendation is one for which the GDG was confident that the desirable effects of adhering to the recommendation outweigh the undesirable effects.

A weak or conditional recommendation is one for which the GDG concluded that the desirable effects of adhering to the recommendation probably outweigh the undesirable effects, but was not confident about these trade-offs.

2.6 Deciding upon recommendations

The GDG met virtually for four sessions in February 2021. Systematic reviews and GRADE tables were presented at the meeting. Formulation of recommendations and their strength were facilitated by the chair and supported by the methodologist, with the aim of achieving unanimous consensus. While the plan was to use a simple majority vote, full consensus was reached on all recommendations.

The GDG used evidence-to-decision tables to guide the process of developing recommendations. These tables addressed the certainty of evidence, the balance between desirable and undesirable effects, values, resource use and cost-effectiveness, equity, acceptability and feasibility. These tables are available in Web Annex B, and the full evidence profiles in Web Annex A. The WHO Steering Group noted remarks made by members of the External Review Group and considered incorporating these into the final guideline. Some evidence-to-decision frameworks were consolidated to provide recommendations that are more practical and implementable from an end-user perspective; hence, the 11 questions led to eight recommendations.

2.7 Funding

The development of this guideline was financially supported by the US Centers for Disease Control and Prevention and the World Health Organization.

3 Recommendations

3.1 Blood pressure threshold for initiation of pharmacological treatment

1. RECOMMENDATION ON BLOOD PRESSURE THRESHOLD FOR INITIATION OF PHARMACOLOGICAL TREATMENT

WHO recommends initiation of pharmacological antihypertensive treatment of individuals with a confirmed diagnosis of hypertension and systolic blood pressure of ≥ 140 mmHg or diastolic blood pressure of ≥ 90 mmHg.

Strong recommendation, moderate- to high-certainty evidence

WHO recommends pharmacological antihypertensive treatment of individuals with existing cardiovascular disease and systolic blood pressure of 130–139 mmHg.

Strong recommendation, moderate- to high-certainty evidence

WHO suggests pharmacological antihypertensive treatment of individuals without cardiovascular disease but with high cardiovascular risk, diabetes mellitus, or chronic kidney disease, and systolic blood pressure of 130–139 mmHg.

Conditional recommendation, moderate- to high-certainty evidence

Implementation remarks:

- Initiation of pharmacological hypertension (HTN) treatment should start no later than four weeks following diagnosis of HTN. If blood pressure level is high (e.g. systolic ≥ 160 mmHg or diastolic ≥ 100 mmHg) or there is accompanying evidence of end organ damage, initiation of treatment should be started without delay.

Evidence and rationale

The GDG reviewed evidence from 14 relevant systematic reviews that summarized data from a large number of randomized trials involving over 120 000 adult participants (Web Annex A). Evidence summaries are presented for the general population and for higher-risk populations (with diabetes (DM), coronary artery disease (CAD), prior stroke) and were presented for various systolic blood pressure (SBP) thresholds (Web Annex A).

The anticipated benefits of a lower blood pressure (BP) target (140 SBP in the general population and 130 SBP in a high-risk population) were reduction in mortality, cardiovascular mortality, stroke, myocardial infarction (MI) and heart failure events. The anticipated harms were mostly not serious side-effects, and some were a surrogate outcome, such as rise in creatinine that may not be clinically relevant. On average, treatment was associated with a reduction in deaths and cardiovascular events that ranged from 5 to 10/1000 and harms that ranged from 20 to 30/1000. The benefits were a reduction in severe events with significant morbidity and mortality whereas the harms were mostly not clinically significant.

In summary, the anticipated benefits were large and clearly outweighed the harms. The overall certainty varied from moderate to high, depending on the BP level and agent used.

Evidence-to-decision considerations

The value of antihypertensive therapy is well accepted by most patients, health care providers, health systems, professional societies and government agencies. From a patient's perspective, preventing cardiovascular events is highly valued. However, some individuals who are eligible for antihypertensive treatment may not present to care, may be lost to follow up, or are prescribed a treatment but fail to take/adhere to the treatment. Treatment may be perceived as low value from the perspective of an asymptomatic patient unless the person is convinced of a trade-off between immediate inconvenience/side-effects and potential long-term health gains (6). The patient perception of an unfavourable cost-benefit may be further exacerbated by the requirement for out-of-pocket payment for appointments or medications. Therefore, the GDG considered that although there is important variability in stakeholder values, overall initiation of hypertension (HTN) medications is likely to be feasible and acceptable overall. Given that the barriers to accessing HTN care in low-income settings include low patient health literacy, lack of financial protections, and limited resources (7), the GDG felt that health inequalities would probably be reduced by HTN treatment.

In terms of costs and resource requirements, the GDG acknowledged variability, based on the structure of a country's public health system and its economic status. Other costs, including human resources and medications, were considered moderate, given the benefits. Multiple sources of cost effectiveness are available from various countries, such as Argentina, Nigeria, the USA and UK (8, 9, 10, 11, 12, 13), and for lower thresholds and higher-risk individuals (14, 15, 16). Most cost-effectiveness estimates were clustered below USD 1000 per averted disability-adjusted life year (DALY) – well below the average 2017 GDP per capita for lower-middle income countries of USD 2188 (17), suggesting that pharmacological treatment could be very cost-effective for this group of countries. HTN treatment (treating all with BP $\geq 140/90$ mmHg) has been shown to be cost-effective and a “best buy” intervention by the Kostova study (8). Treating high-risk/CVD patients with baseline 130–139 mmHg has been shown to be cost effective, but not cost saving (SPRINT cost-effective analyses) (15); value depends on maintaining the intervention effect for more than five years.

3.2 Laboratory testing before and during pharmacological treatment

2. RECOMMENDATION ON LABORATORY TESTING

When starting pharmacological therapy for hypertension, WHO suggests obtaining tests to screen for comorbidities and secondary hypertension, but only when testing does not delay or impede starting treatment.

Conditional recommendation, low-certainty evidence

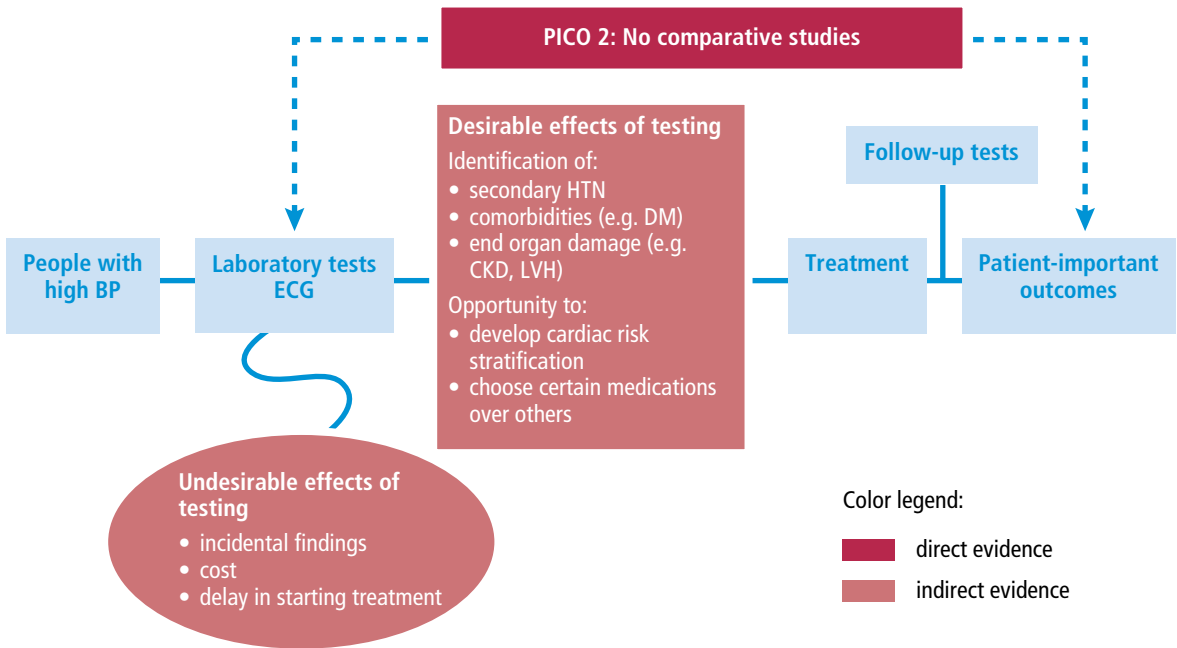
Implementation remarks:

- Suggested tests include serum electrolytes and creatinine, lipid panel, HbA1C or fasting glucose, urine dipstick, and electrocardiogram (ECG).
- In low-resourced areas or non-clinical settings, where testing may not be possible because of additional costs, and lack of access to laboratories and ECG, treatment should not be delayed, and testing can be done subsequently.
- Some medicines, such as long-acting dihydropyridine calcium-channel blockers (CCBs) are more suitable for initiation without testing, compared to diuretics or angiotensin-converting enzyme inhibitors (ACEi)/angiotensin-II receptor blockers (ARBs).

Evidence and rationale

Comparative studies that evaluate various testing strategies prior to the initiation of antihypertensive treatment were not identified, despite a search of the literature. Therefore, indirect evidence was sought to evaluate this question. An analytic framework was developed to conceptualize the rationale for obtaining diagnostic testing, such as laboratory tests and electrocardiogram (ECG), in this context (see Fig. 2). This framework identified the four most important reasons for obtaining testing, which are to diagnose secondary HTN, identify comorbidities (e.g. DM), identify end organ damage (e.g. chronic kidney disease (CKD) or left ventricular hypertrophy (LVH)), and for cardiac risk stratification.

Fig. 2 Framework for analysis



In terms of secondary HTN, various studies suggest a prevalence of 5–10% among patients with a diagnosis of HTN and a higher prevalence of 10–30% among patients with particularly high BP (e.g. over 175/115 mmHg) or BP that is resistant to treatment (18, 19, 20). Morbidities and end organ damage that could be identified by testing patients with HTN are also common. One estimation indicates that 23%, 24% and 39% of patients with a diagnosis of HTN have one, two, three or more comorbidities respectively. Common comorbidities among patients with HTN that can be discovered via laboratory testing are hyperlipidaemia and diabetes, which have a prevalence of 56% and 27% respectively (21). Testing at the point of starting HTN medications or for subsequent monitoring can also identify patients who develop certain adverse events after treatment (e.g. hyperkalaemia and acute kidney injury), thus providing a rationale for testing. Testing also had the additional advantage of identifying compelling indications for choosing certain medications over others. For example, identifying diabetes would favour the use of ACEis/ARBs, and identifying hyponatraemia would lead to not starting diuretics. Overall, the desirable effects of testing were judged to be at least moderate.

The framework also identified the most undesirable effects of obtaining testing, which were delay of starting treatment, cost and incidental findings. Delay of treatment was judged to be the most important concern since it can lead to losing the patient for follow up and the potential for adverse cardiovascular (CV) outcomes. Incidental findings on testing were thought to be less important. The undesirable effects of testing were judged to have smaller magnitude. On balance, desirable effects are likely to outweigh undesirable effects. The certainty of evidence across outcome was judged to be very low due to serious concerns about the indirectness of evidence.